

CH1101: Quantum Chemistry and Spectroscopy

Origins of Quantum Theory

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Bohr's theory of hydrogen atom

- ▶ Radius of Bohr orbit
- ▶ Energy of Bohr orbit



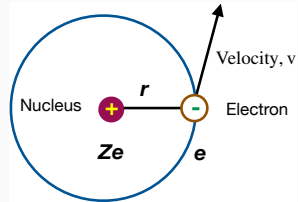
I. Radius of Bohr orbit

Coulombic force:

$$f_{\text{Coulomb}} = \frac{ze \times e}{4\pi\epsilon_0 r^2}$$

Centrifugal force:

$$f_{\text{Centrifugal}} = \frac{mv^2}{r}$$



Assumption 1: Existence of stationary circular orbit

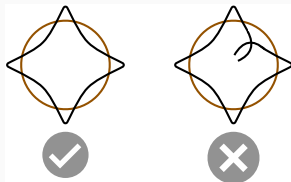
$$f_{\text{Coulomb}} = f_{\text{Centrifugal}} \implies \frac{ze \times e}{4\pi\epsilon_0 r^2} = \frac{mv^2}{r} \quad (1)$$

I. Radius of Bohr orbit

Assumption 2: Only a few specified orbits allowed

$$2\pi r = n\lambda \implies \frac{2\pi r}{n} = \lambda \quad (2)$$

$2\pi r$ is the circumference of the orbit;
 $n = 1, 2, 3, \dots$



Wavelength of orbiting electron must “match” or be in phase after making a complete revolution

I. Radius of Bohr orbit

As de Broglie's wave-particle duality states:

$$\begin{aligned}\lambda &= \frac{h}{p} \\ \Rightarrow \frac{2\pi r}{n} &= \frac{h}{p} = \frac{h}{mv} \\ \Rightarrow v &= \frac{nh}{2\pi rm}\end{aligned}\tag{3}$$

Substituting v into Eq. 1

$$\begin{aligned}\frac{ze \times e}{4\pi\epsilon_0 r^2} &= \frac{m\left(\frac{nh}{2\pi rm}\right)^2}{r} \\ \Rightarrow \frac{ze^2}{4\pi\epsilon_0 r} &= m\left(\frac{nh}{2\pi rm}\right)^2 \\ \Rightarrow \frac{ze^2}{4\pi\epsilon_0} &= \frac{n^2 h^2}{4\pi^2 rm} \\ \Rightarrow r &= \frac{4\pi\epsilon_0 n^2 h^2}{ze^2 4\pi^2 m} = \frac{4\pi\epsilon_0 n^2 \hbar^2}{ze^2 m} = \frac{\epsilon_0 n^2 h^2}{ze^2 \pi m}\end{aligned}\tag{4}$$

I. Radius of Bohr orbit

$$r = \frac{\epsilon_0 n^2 h^2}{ze^2 \pi m}$$

As $n = 1, 2, 3, \dots$ Bohr radius is dependent on certain discrete values of n . Therefore, we can simply say that **Bohr orbit is quantized!**

- Problem: Find the radius of 1st Bohr orbit in Å unit for H-atom



II. Energy of Bohr orbit

The total energy (E) of the electron of an atom is sum of its kinetic energy (T) and potential energy (V)

$$\begin{aligned} E &= T + V \\ &= \frac{1}{2}mv^2 + \left(-\frac{ze \times e}{4\pi\epsilon_0 r}\right) \end{aligned}$$

Using Eq. 1

$$\begin{aligned} E &= \frac{1}{2} \frac{ze \times e}{4\pi\epsilon_0 r} + \left(-\frac{ze \times e}{4\pi\epsilon_0 r}\right) \\ &= -\frac{1}{2} \left(\frac{ze^2}{4\pi\epsilon_0 r}\right) \\ &= -\frac{ze^2}{8\pi\epsilon_0 r} \end{aligned}$$



II. Energy of Bohr orbit

Substituting r from Eq. 4

$$\begin{aligned} E &= -\frac{ze^2}{8\pi\epsilon_0\left(\frac{\epsilon_0 n^2 h^2}{ze^2 \pi m}\right)} \\ &= -\left(\frac{ze^2 \times ze^2 \pi m}{8\pi\epsilon_0 \times \epsilon_0 n^2 h^2}\right) \\ &= -\frac{z^2 e^4 m}{8\epsilon_0^2 n^2 h^2} \end{aligned} \tag{5}$$

Here, $n = 1, 2, 3, \dots$ indicating **energy of electron is quantized!**

$$E = -\frac{z^2 e^4 m}{8\epsilon_0^2 n^2 h^2}$$

Theory	Position	Momentum
Bohr	✓	✓
Heisenberg	✓	✗
Heisenberg	✗	✓